

Upper-Level Distinctions + Foundational Ontologies

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Ontology Library (*final*)

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Agenda

- Applied Ontology (AO)
 - What is Applied Ontology?
 - AO History
 - What Are Ontologies for?
 - Interdisciplinarity of AO
- Ontology & -gies
 - What is (an) Ontology?
 - Types of Ontologies
 - What is Formal Ontological Analysis? + OntoClean
- Upper-Level Distinctions + Foundational Ontologies
 - TLO Overview
 - **repeat**
 - Distinctions
 - Comparisons
 - **until** exhausted
 - Questions

Formal Tools for Ontological Analysis

per each item: **the theory of**

- parts (mereology)
- unity and plurality
- **essence** and **identity** →
- dependence
- composition and constitution
- properties and qualities
- time, persistence and change

→ cf. „Mereology“ on Tue

- ,interconnections of things‘ ex.:
if $a = b$ and $b = c$, then $a = c$
(irrespective what a-c are)
- like logic: implication-transitivity

→ cf. „Time, ... Events“ on Tue

Cleaning Taxonomies

Concepts, Attributes, and Arbitrary Relations
Some Linguistic and Ontological Criteria
for Structuring Knowledge Bases

1992

- use of formal ontological tools
from ≈ 1992 on

- example definitions: **founded + rigid** $\rightarrow$ **role**

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2.1. The notion of foundation

In order to be able to propose a more adequate condition, we have first to introduce the notion of *foundation* among concepts, typical of Husserl's ontology [20]. Foundation has been formalized in [18] by the modal formula below, where $\in$ is the membership relation¹, $\leq$ the part-of relation and **nec** stands for the modal necessity (box) operator:

Definition 2.1. The concept α is *founded on* β' (written $\alpha \Downarrow \beta$) if:

$$\mathbf{nec} \forall_x (x \in \alpha \supset \exists y (y \in \beta \wedge x \not\leq y \wedge y \not\leq x)) .$$

α is called *founded* (written $\alpha \Downarrow$) if there exists a β such that $\alpha \Downarrow \beta$. α is called *essentially independent* (written $I(\alpha)$) if $\neg(\alpha \Downarrow)$, and *self-founding* if $\alpha \Downarrow \alpha$.

Cleaning Taxonomies

Concepts, Attributes, and Arbitrary Relations Some Linguistic and Ontological Criteria for Structuring Knowledge Bases

1992

- use of formal ontological tools
from ≈ 1992 on

- example definitions: **founded + not rigid $\rightarrow$ role**

2.1. The notion of

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been formalized in
the part-of relatic

Definition 2.1. Th

$\text{nec } \forall_x (x$

α is called *founde*
independent (writ

Definition 2.2. A concept α is called *semantically rigid* (written $R(\alpha)$) if

$$\forall_x (x \in \alpha \supset \text{nec}(x \in \alpha)) .$$

Intuitively, we can think that a concept α is semantically rigid if it contributes to the very identity of its instances, in such a way that, if α is an α in a particular situation, it has to keep to be an α in any possible situation in order to keep its identity. For instance, an animal can cease to be a pup while still being a dog: *animal* and *dog* are semantically rigid, *pup* is not.

Definition 2.3. A concept α is called a *role* if it is founded but not semantically rigid, that is,
 $\alpha \Downarrow \wedge \neg R(\alpha)$.

Cleaning Taxonomies: OntoClean

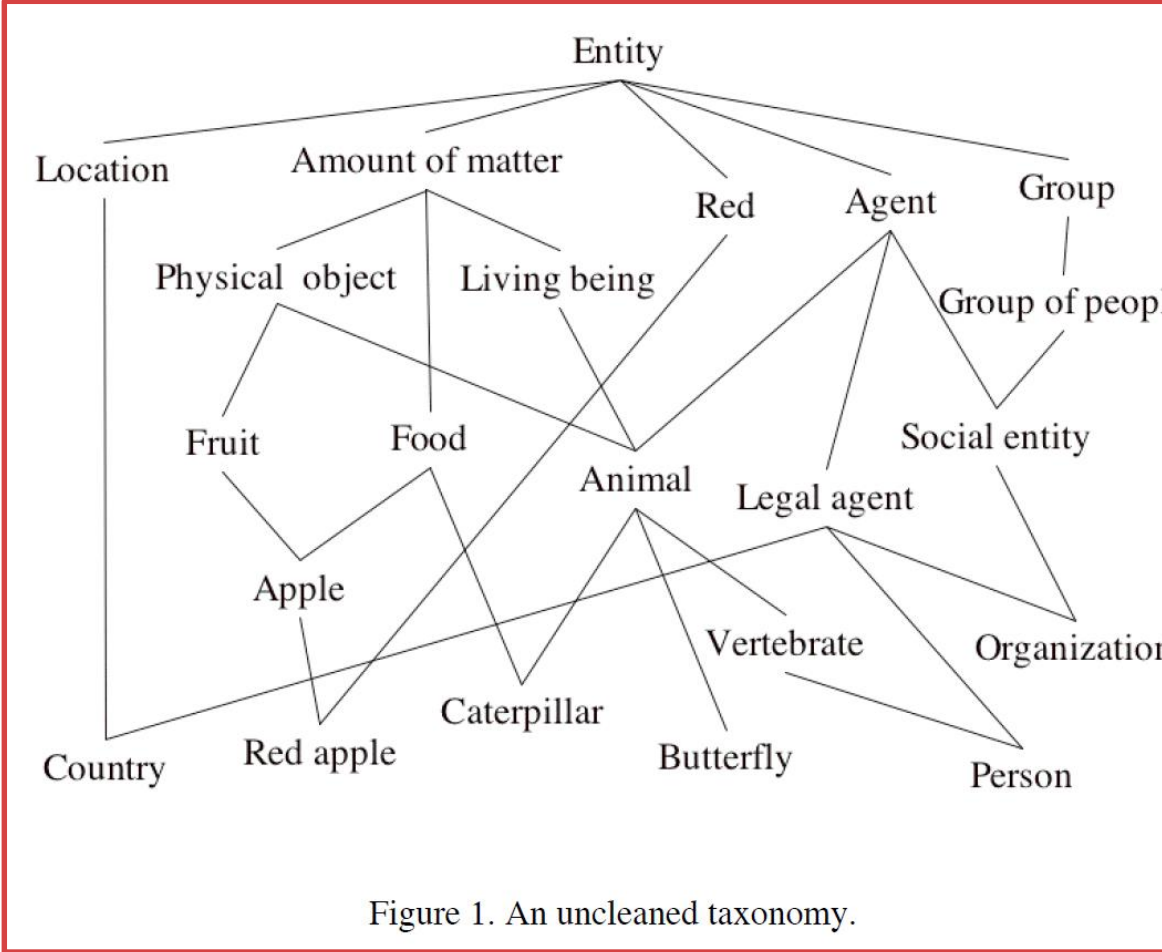


Figure 1. An uncleaned taxonomy.

[Guarino & Welty, 2001 & 2004]

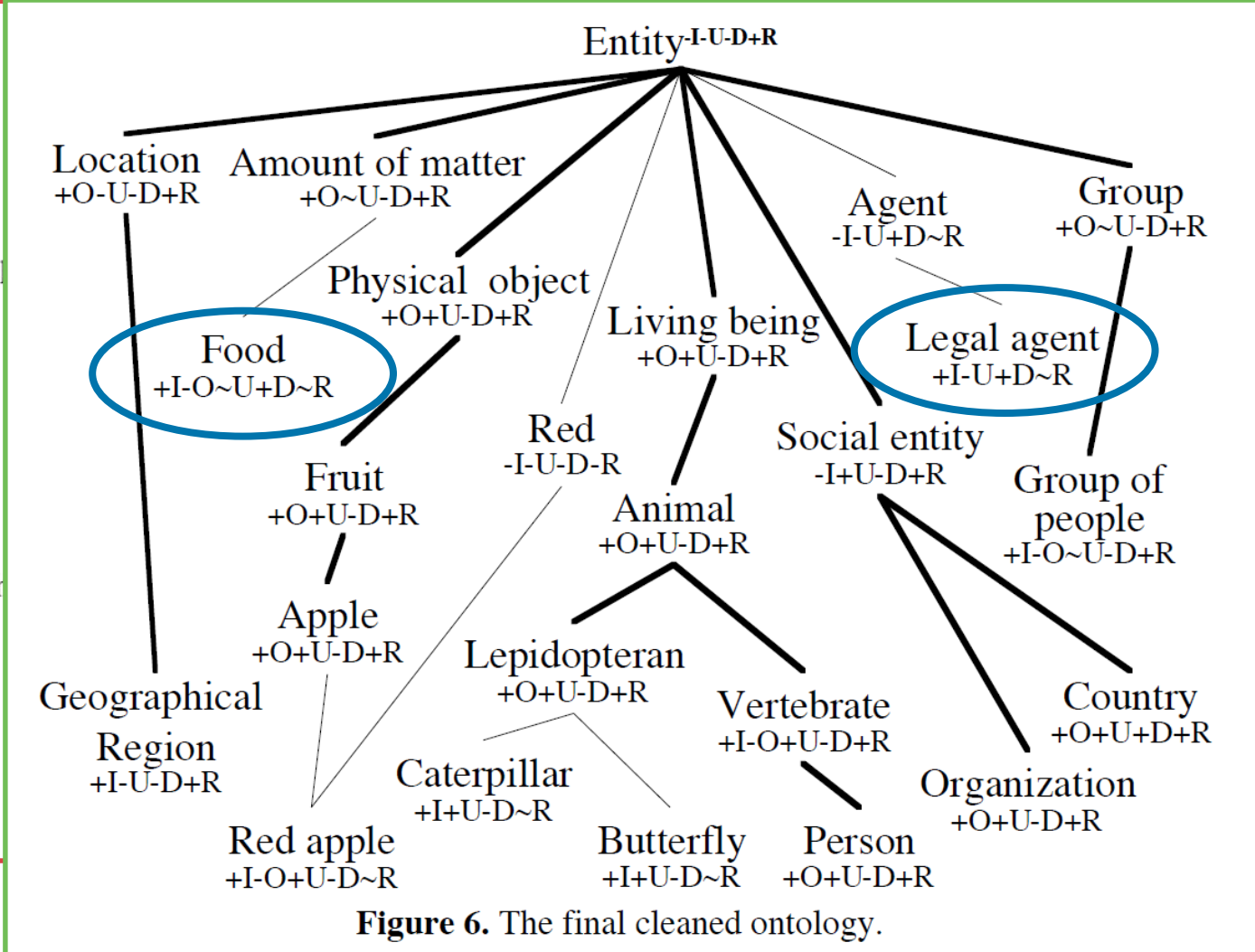


Figure 6. The final cleaned ontology.

Cleaning Taxonomies: OntoClean

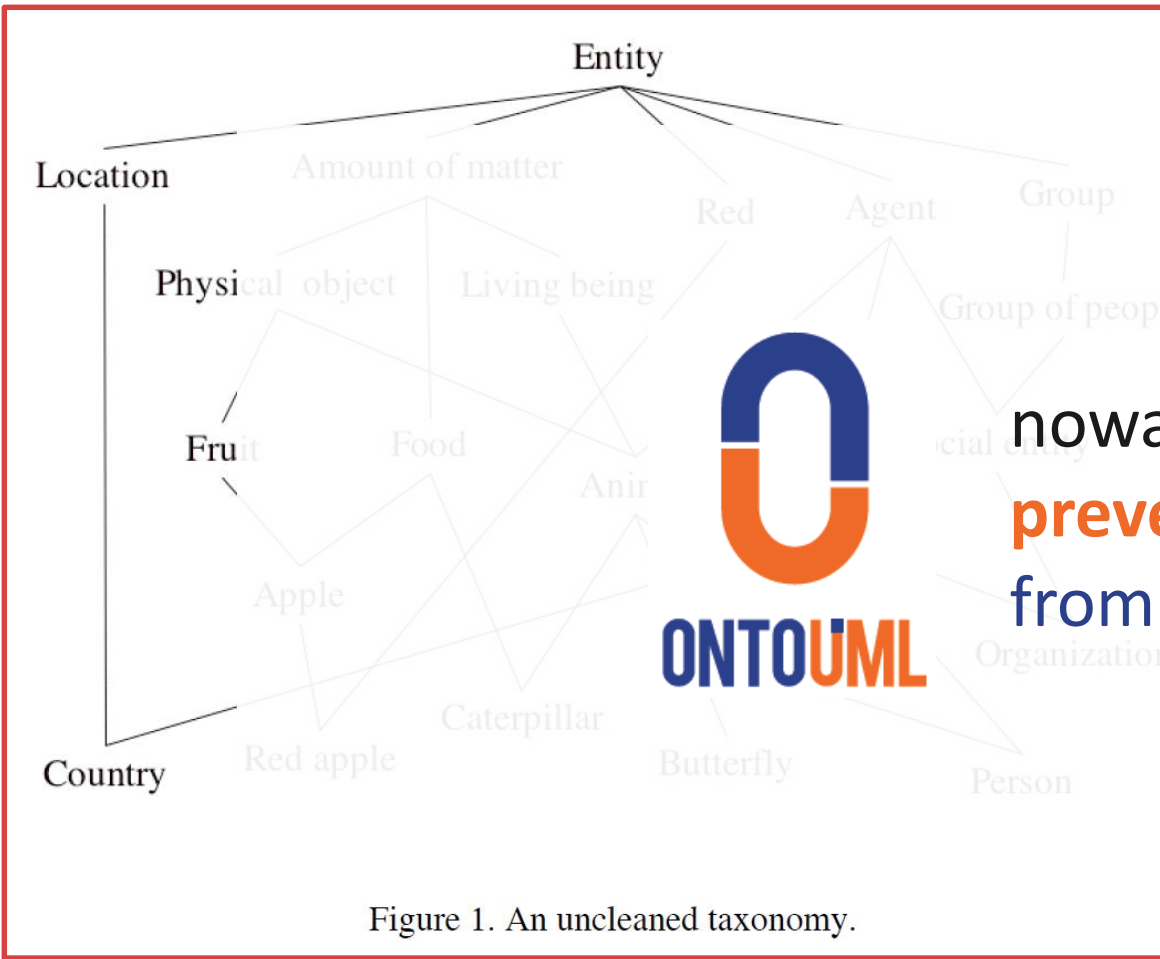


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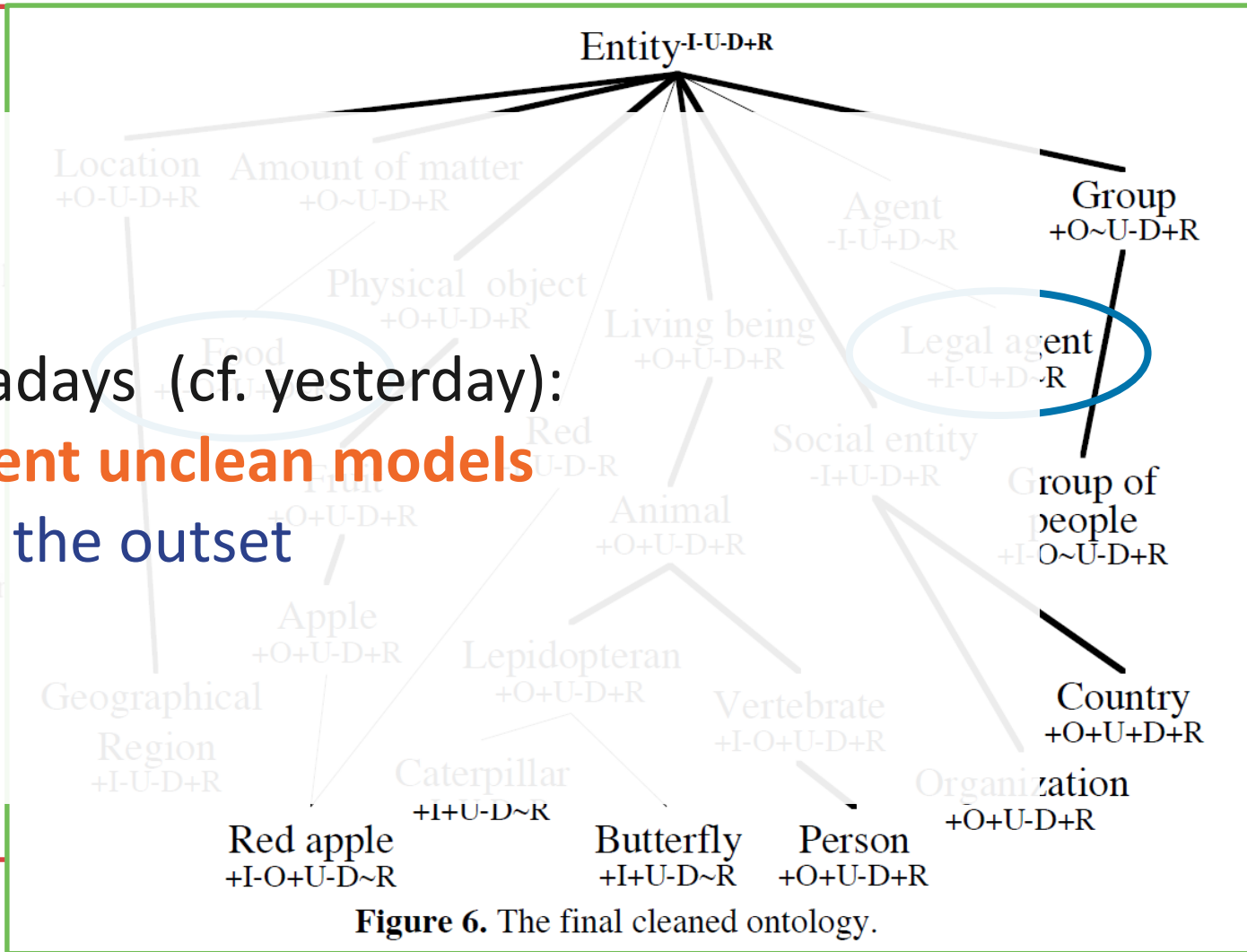


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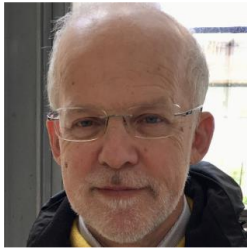
Distinctions + Foundational Ontologies

aka top-level ontologies (TLOs)

TLO Selection

selected

- BFO Basic Formal Ontology B. Smith
- DOLCE Descriptive Ontology of Cognitive and Linguistic Engineering N. Guarino & LOA
- UFO Unified Foundational Ontology G. Guizzardi
- [GFO General Formal Ontology H. Herre]



others in the AO-2022 special issue on TLOs

- GUM Generalized Upper Model J. Bateman
- TUpper [?Toronto Upper?] M. Gruninger
- YAMATO Yet Another More Advanced Top-Level Ontology R. Mizoguchi

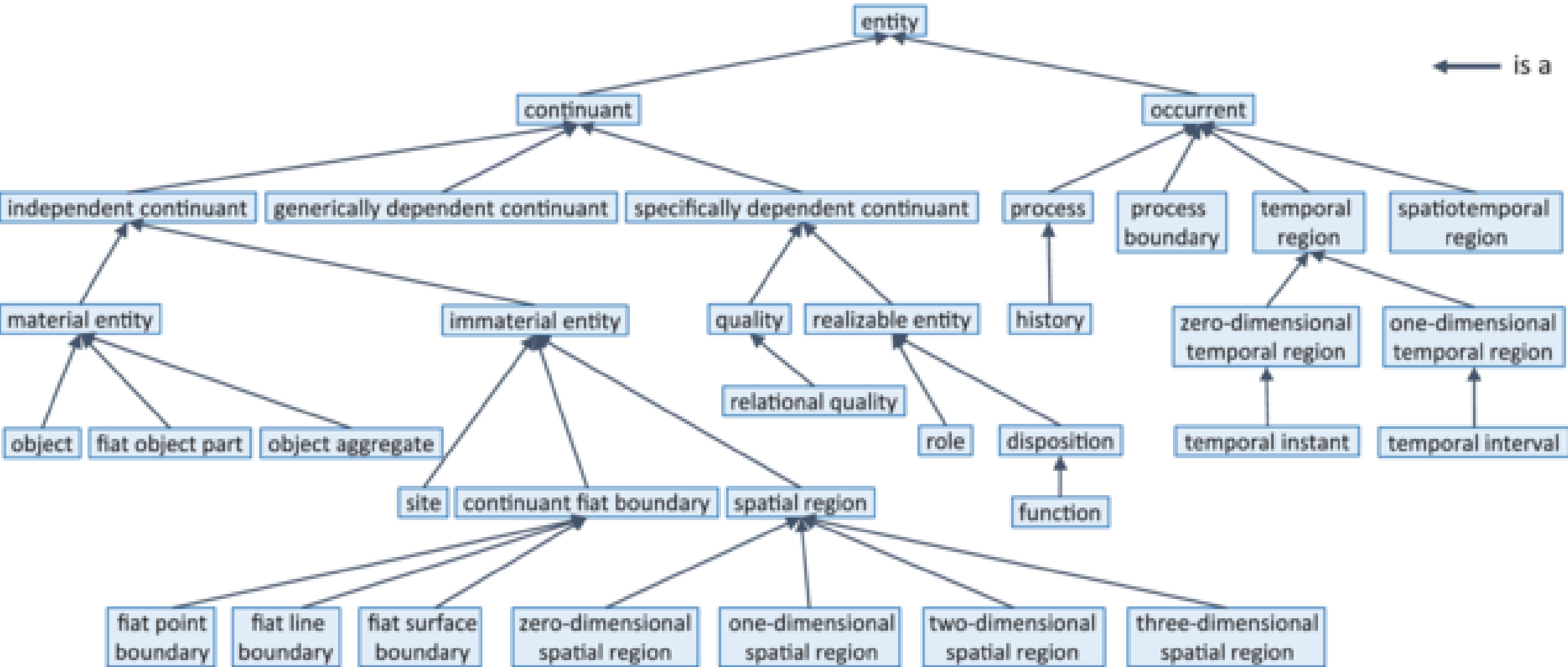
others likely active

- BORO Business Object Reference Ontology C. Partridge
- CYC (its upper level) Cycorp; D. Lenat[†]
- EMMO Elementary Multiperspective Material Ontology E. Ghedini
- gist gist D. McComb, M. Uschold
- SUMO Suggested Upper Merged Ontology A. Pease
- ?ThingFO Thing Foundational Ontology? L. Olsina

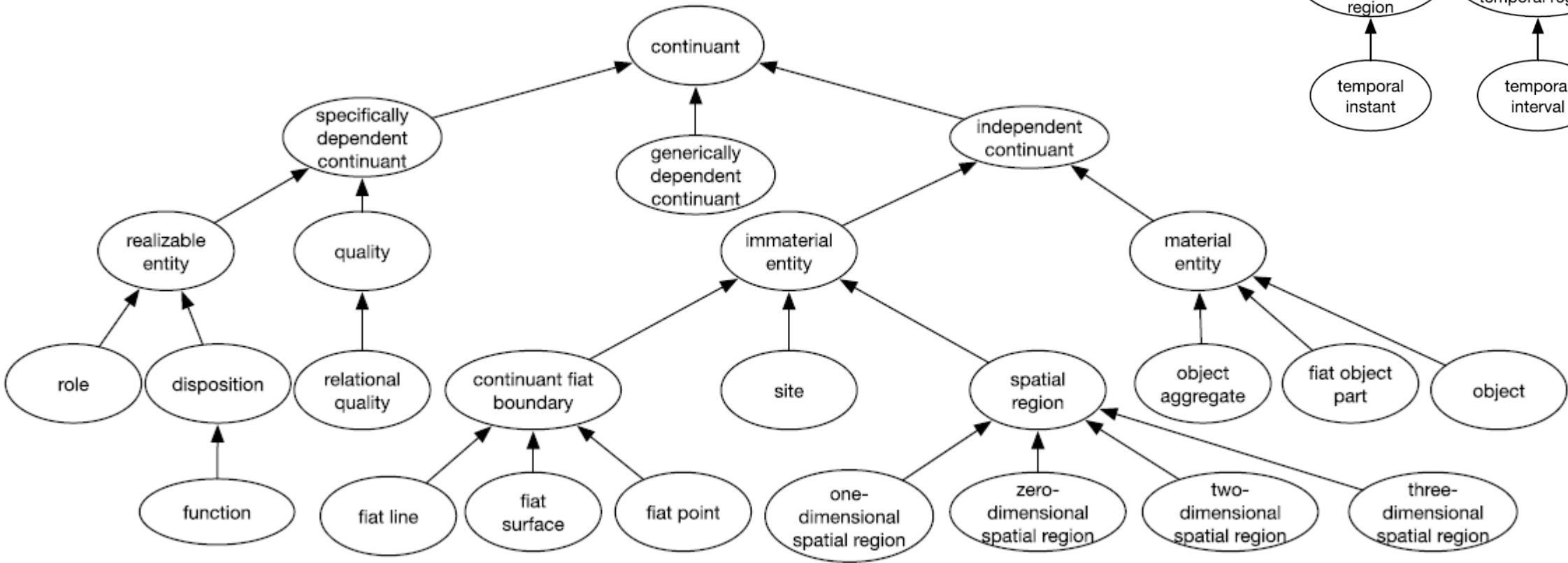
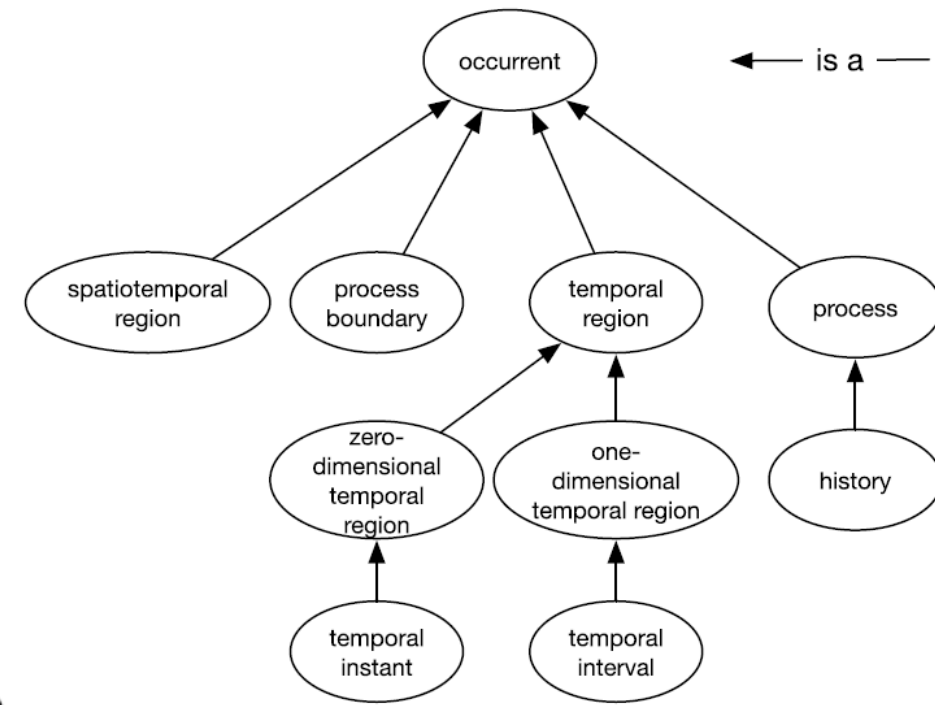
no news for a while

- BWW Bunge-Wand-Weber Y. Wand, R. Weber
- ?COSMO Common Semantic Model? P. Cassidy
- IEEE SUO/IFF Standard Upper Ontology/Information Flow Framework R. Kent
- ISO 15926 Integration of Lifecycle Data... M. West
- J. Sowa's (from his KR book, 2000)
- SENSUS/HPKB ?/High Performance Knowledge Bases ?S. Aitken?

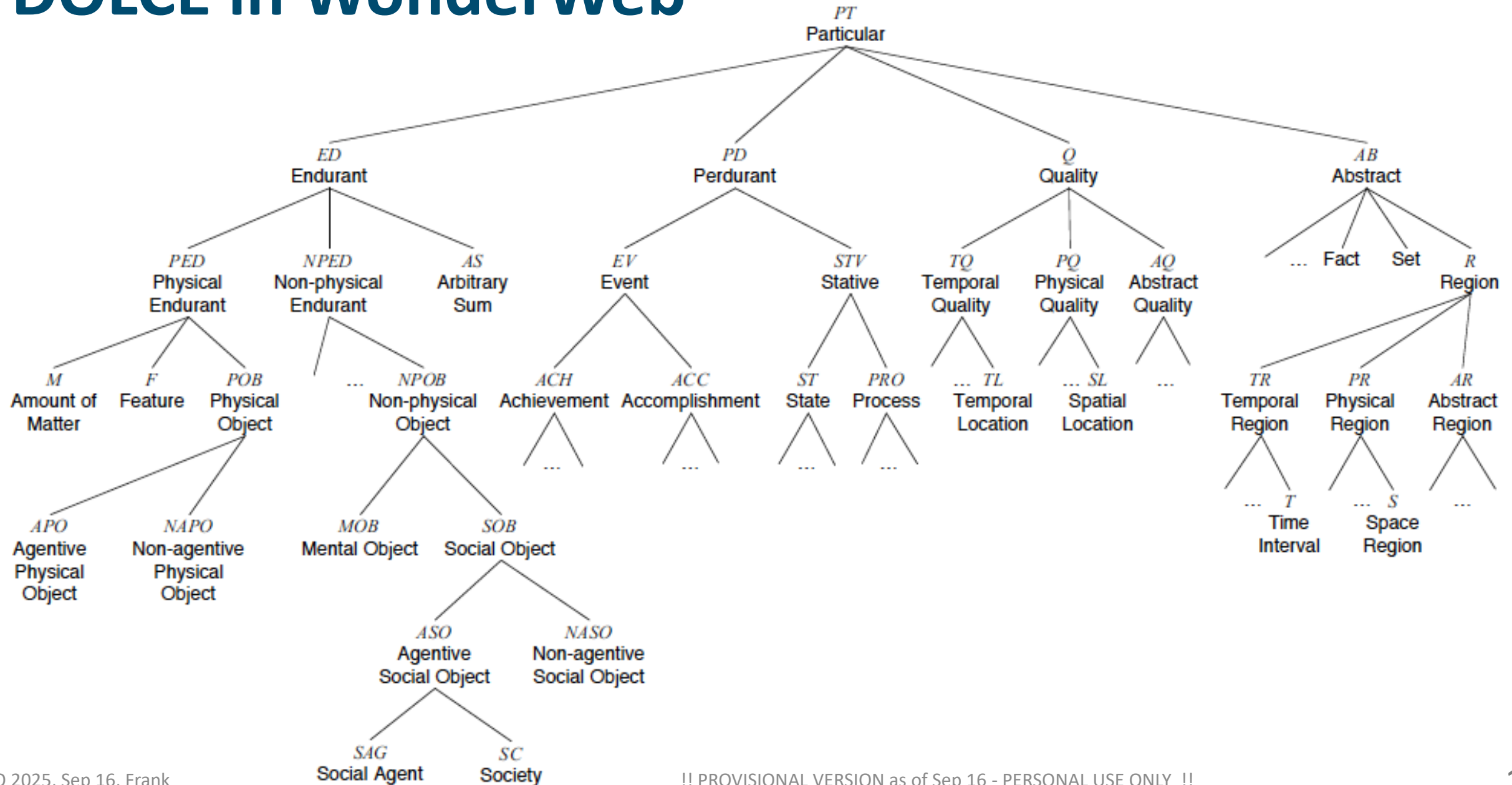
BFO in ISO-21838-2



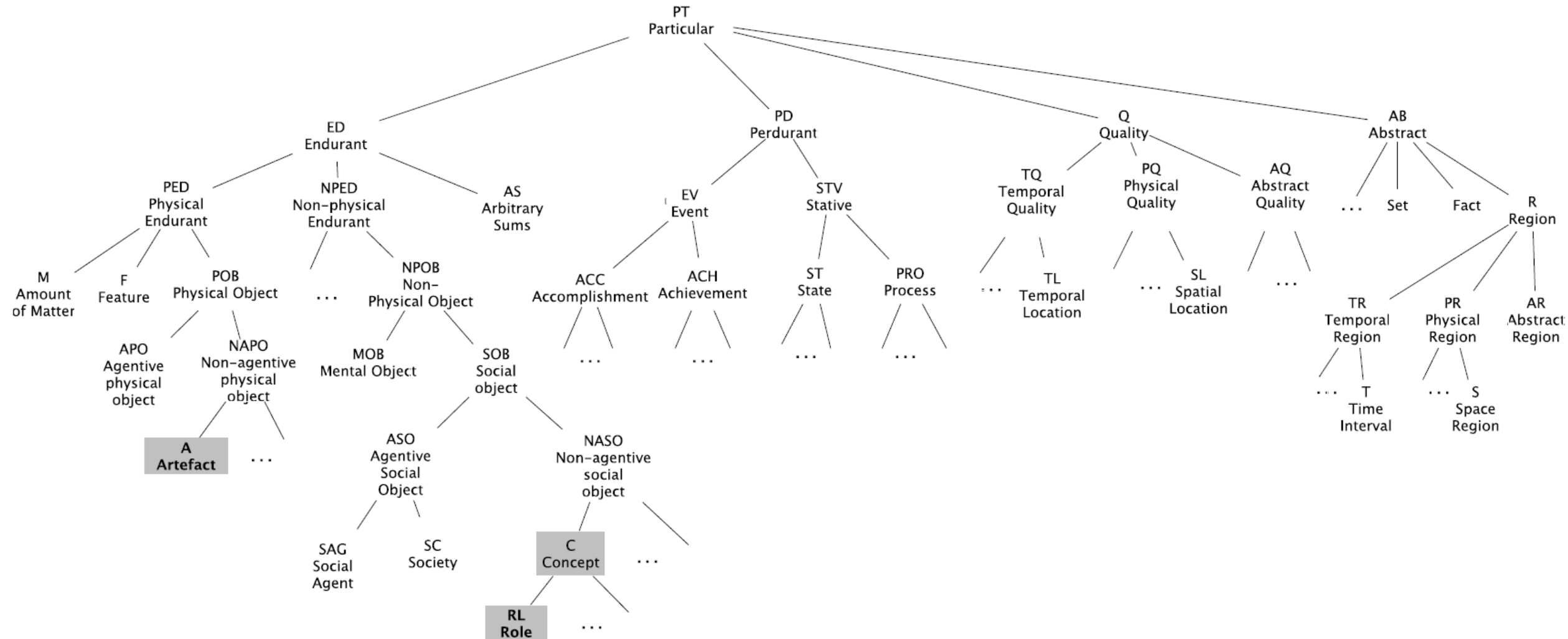
BFO in AO-2022



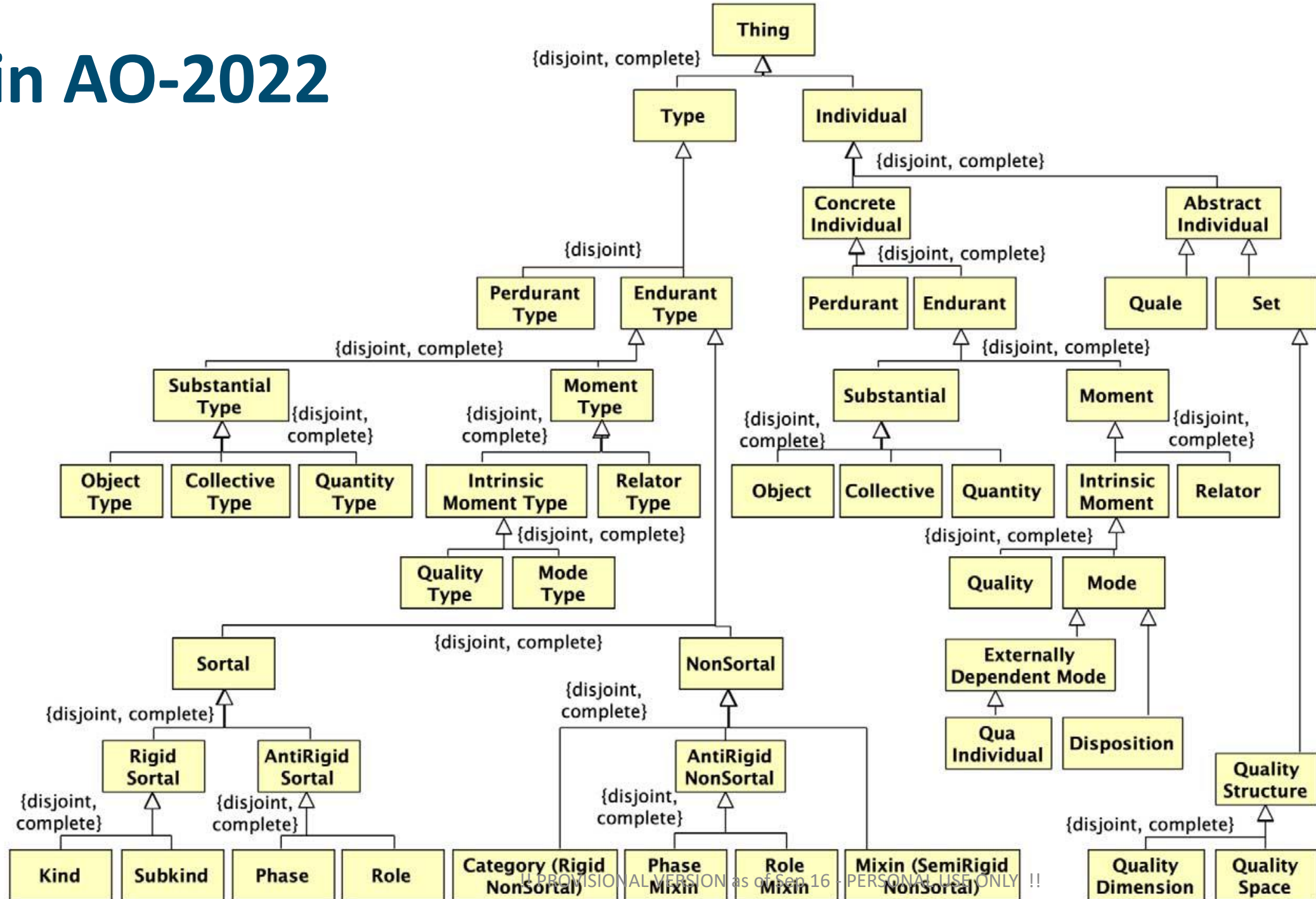
DOLCE in WonderWeb



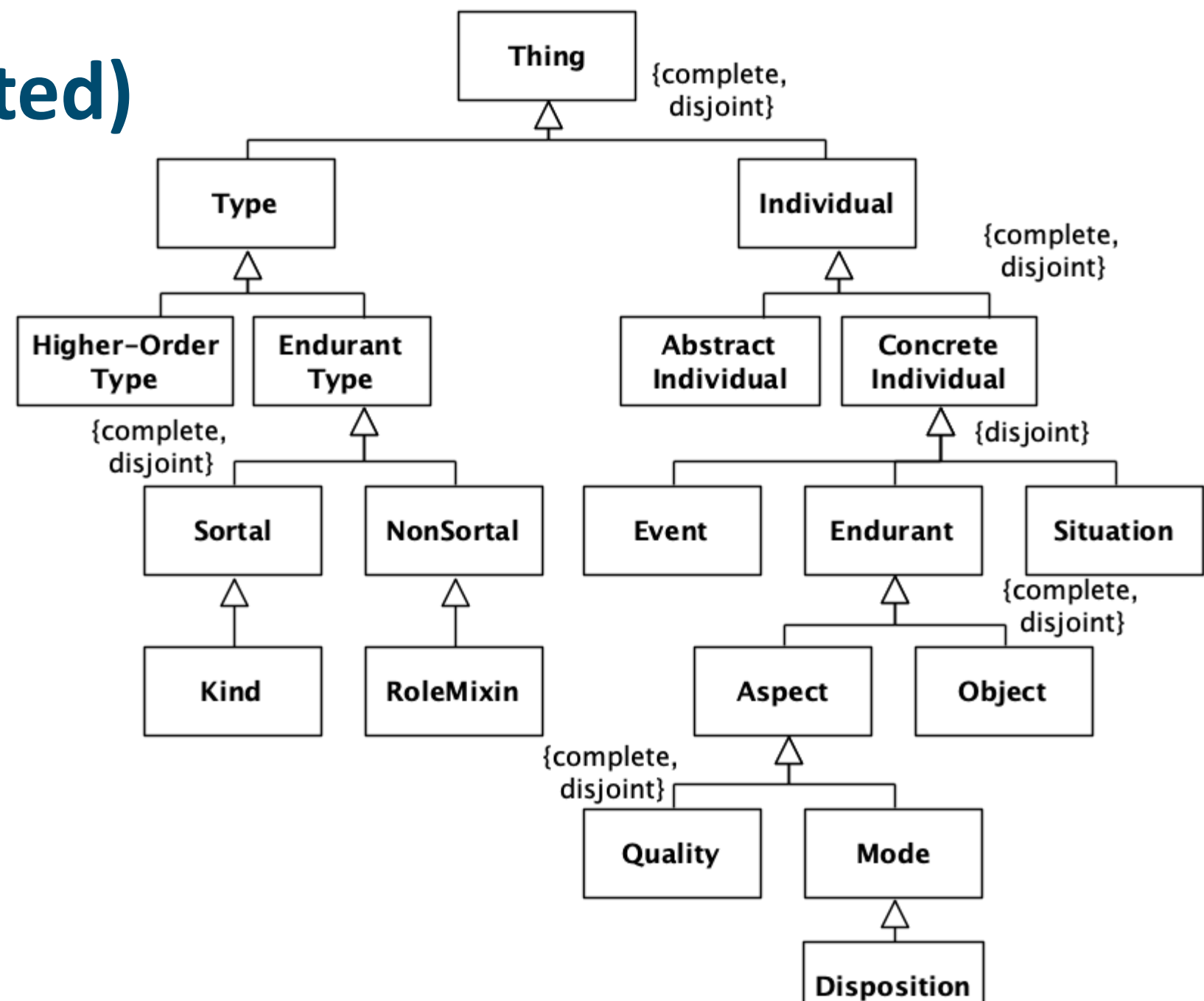
DOLCE in AO-2022



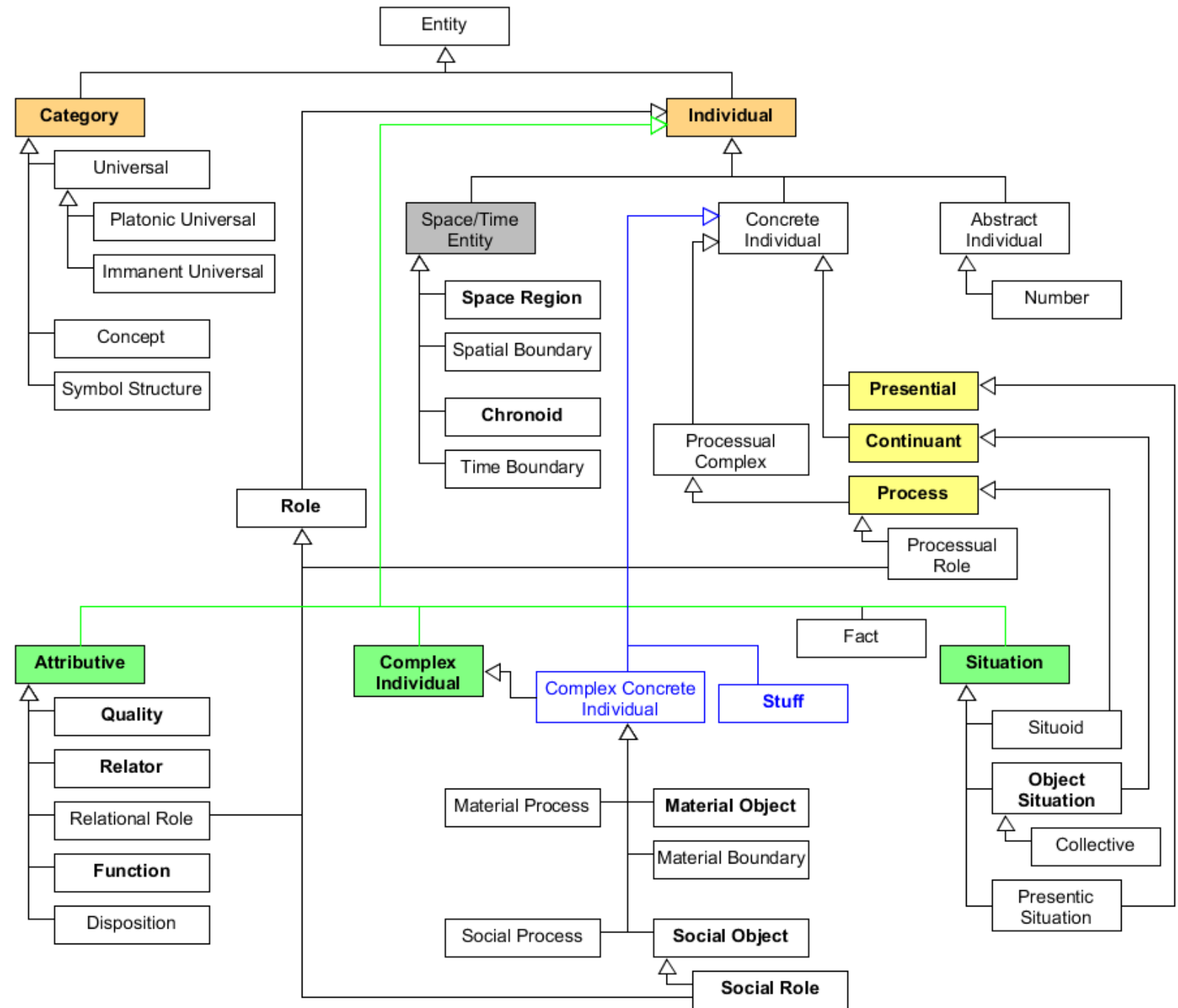
UFO in AO-2022



UFO (selected)



GFO in AO-2022

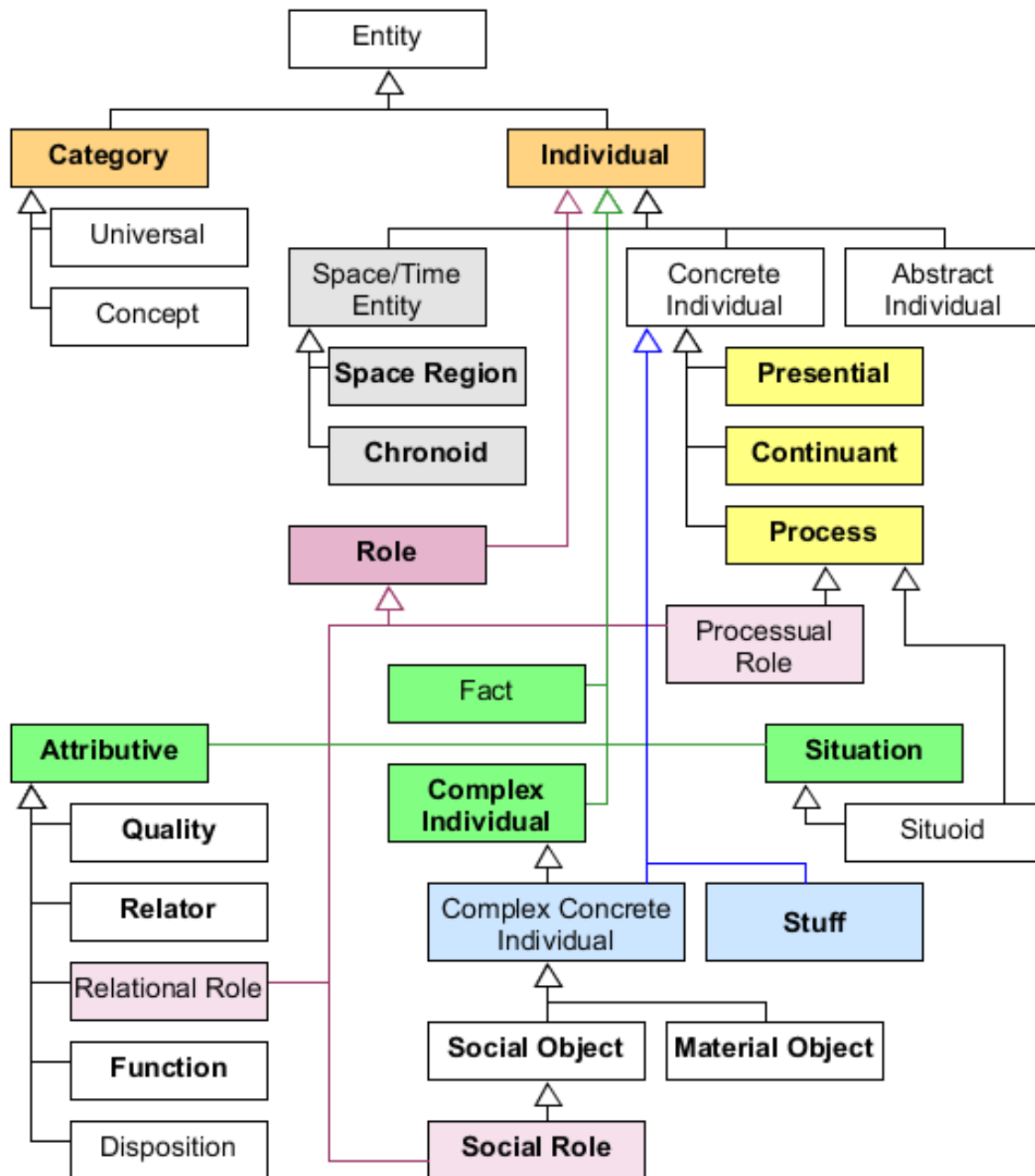


Selected GFO Categories

(2024 taxonomic overview)

highlighting

- categories included
- object-process integration
- complexity of individuals
- amount of matter
- roles (*crosscutting branches*)
- space and time



Type – Individual

- e.g. *homo sapiens* – I myself
- based on *instantiation* $(x :: t)$
[rdf:type]
- individuals cannot be instantiated
- higher-order types are a choice
e.g. *species* – *homo sapiens*
- remarks
 - *extensional* subsumption is defined via instantiation → below
 - *intensional* subsumption depends on the understanding of type
 - ! use of „isA“ for *instanceOf* vs *subTypeOf* !

$$\forall st. \text{subClassOf}(s, t) \leftrightarrow (\forall x. (x :: s \rightarrow x :: t))$$

[rdfs:subClassOf]

Continuant – Occurrent

- e.g. the monastery – this lecture
- continuant
 - wholly present at each time of its existence
 - „wholly“ = all parts of it
 - thus can change/is mutable
- occurrent
 - unfolds/happens over time
 - can be/while unfolding only partially present
 - thus cannot change/immutable

participation links instances of both

Notation / Preliminaries

- prefixes/“namespaces”
 - bfo: , dolce: , ufo: for lines or larger pieces
 - b:, d:, u:, d: within single lines
- “simsets” – constituents deemed quite similar *in themselves*
 - e.g. { b:entity, d:particular }

{b:entity, d:particular, u:individual, g:individual}

{u:thing, g:entity} ≠ b:entity

{b:continuant, d:endurant}

The Top-Level Distinction

- bfo: entity → continuant | occurrent
- dolce: particular → endurant | perdurant | quality | abstract
- ufo: thing → type | individual
- gfo: entity → category | individual

observations

- UFO (+ GFO) have types in taxonomy; BFO, DOLCE in their meta-theory
 - UFO with elaborate type hierarchy and theory of types (as seen yesterday)
- DOLCE with 3 distinctions at once:
(1) endurant-perdurant, (2) quality-substance, (3) abstract-concrete

{b:occurrent, d:perdurant, u:perdurant, g:occurrent}

{b:continuant, d:endurant, u:endurant, g:continuant}

Endurant – Perdurant

- bfo: entity → continuant | occurrent
- dolce: particular → endurant | perdurant | quality | abstract
- ufo: thing → type | individual
- ufo: concrete individual → endurant | perdurant
- gfo: concrete individual → continuant | occurrent | presential

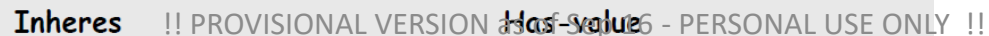
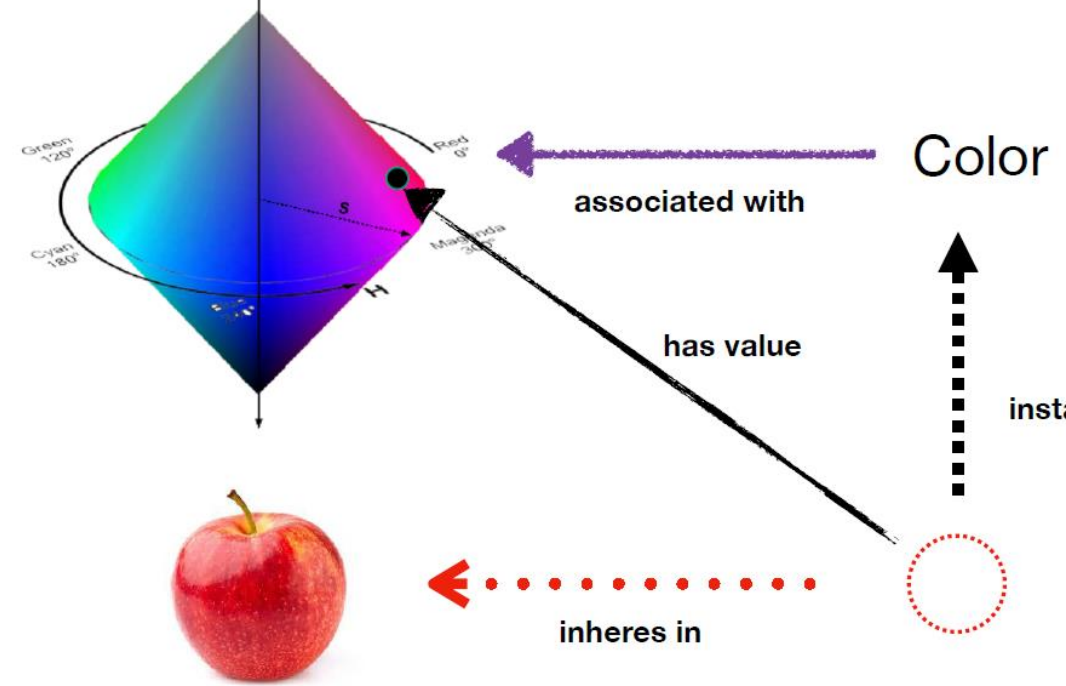
observations

- DOLCE with 3 distinctions at once:
(1) endurant-perdurant, (2) quality-substance, (3) abstract-concrete
- specialty in GFO: presentials – reified ‘snapshots’ at time points

Substance – Quality

- e.g. “an apple A is red”
- options for unpacking this
 - (1) apple A **instantiates** a type ‘to-be-[a-substance-]with-a-specific-redness’
 - (2) a particular quality q **inheres** in apple A and q instantiates redness r, a type of quality
 - (3) a particular quality q **inheres** in apple A and **has value** v, with v being a **part of** a quality region r, thereby **locating** q in r

(3) a particular quality q **inheres** in apple A and **has value** v,
with v being a **part of** a quality region r, thereby **locating** q in r



Ontological Choices

- type-instance, subsumption, higher-order types
- whole-part, unity, boundaries
- substance-quality-quale
- location: multiple objects possible?
- materialism vs abstract objects
- possibilia or not?
- identity defined extensionally vs intensionally
- relational arity: more than 2?
- time: presentist vs eternalist
- indexicals for here and now
- quantities & math entities
- processes and events
- causality
- information & reference
- artefacts & socially constructed entities
- descriptive vs. revisionary; realist vs conceptualist vs nominalist

Questions to TLO specialists

Questions for TLO developers

- Who contributed to the development?
 - rather few vs community-effort?
- Can TLOs be (Husserlian-)formal ontologies?

Brief Wrap

PROVISIONAL VERSION DURING ISAO 2025, FOR PERSONAL USE OF REGISTERED PARTICIPANTS
(as of Sep 16: additions, esp. references of contributions to come)

- all of BFO, DOLCE, UFO
 - are in a well-developed state
 - with their respective user bases
 - ?in the process of ISO standardization?

**Thanks! Happy
about further discussion**

- remember:

